THE UNIVERSITY OF OKLAHOMA

School of Computer Science & Data Science and AI

A JOB-SHOP ACCOUNTING SYSTEM

End-to-End Conceptual, Relational & Cloud DBMS Implementation

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SECTION IV. TABLE OF CONTENTS

Task 1. Entity-Relationship (ER) Diagram 2

Task 2. Relational Database Schemas 3

Task 3. Physical Storage Structures 4
- 3.1. Discussion of Storage Structures for Tables (Lecture Topic 4) . . . 4
- 3.2. Discussion of Storage Structures for Tables (Azure SQL Database) . . . 4

Task 4. SQL Statements & Azure SQL Table Creation 5

Task 5. Query Implementation & Java JDBC Program 6
- 5.1. SQL Statements & Transact-SQL Stored Procedures . . . 6
- 5.2. Java Source Program & Successful Compilation . . . 6

Task 6. Java Program Execution Screenshots 7
- 6.1. Query 1 . . . 7
- 6.2. Query 2 . . . 8
- 6.3. Query 3 . . . 9
- 6.4. Query 4 . . . 10
- 6.5. Query 5 . . . 11
- 6.6. Query 6 . . . 12
- 6.7. Query 7 . . . 13
- 6.8. Query 8 . . . 14
- 6.9. Query 9 . . . 15
- 6.10. Query 10 . . . 15
- 6.11. Query 11 . . . 16
- 6.12. Query 12 . . . 16
- 6.13. Query 13 . . . 17
- 6.14. Query 14 . . . 17
- 6.15. Options 15 & 16: Import and Export . . . 18
- 6.16. Error Testing . . . 19
- 6.17. Option 17: Exit . . . 19

Task 1. Entity-Relationship (ER) Diagram

The ER diagram models the Job-Shop Accounting System in Silberschatz Chen notation. Rectangles represent entities, diamonds denote relationships, triangles represent ISA specializations, and ovals contain attributes with primary keys underlined.

Process, Job, and Account hierarchies use disjoint (d) partial specialization. The Job entity directly links an Assembly and a Process, and Cost Transactions update corresponding accounts when recorded.

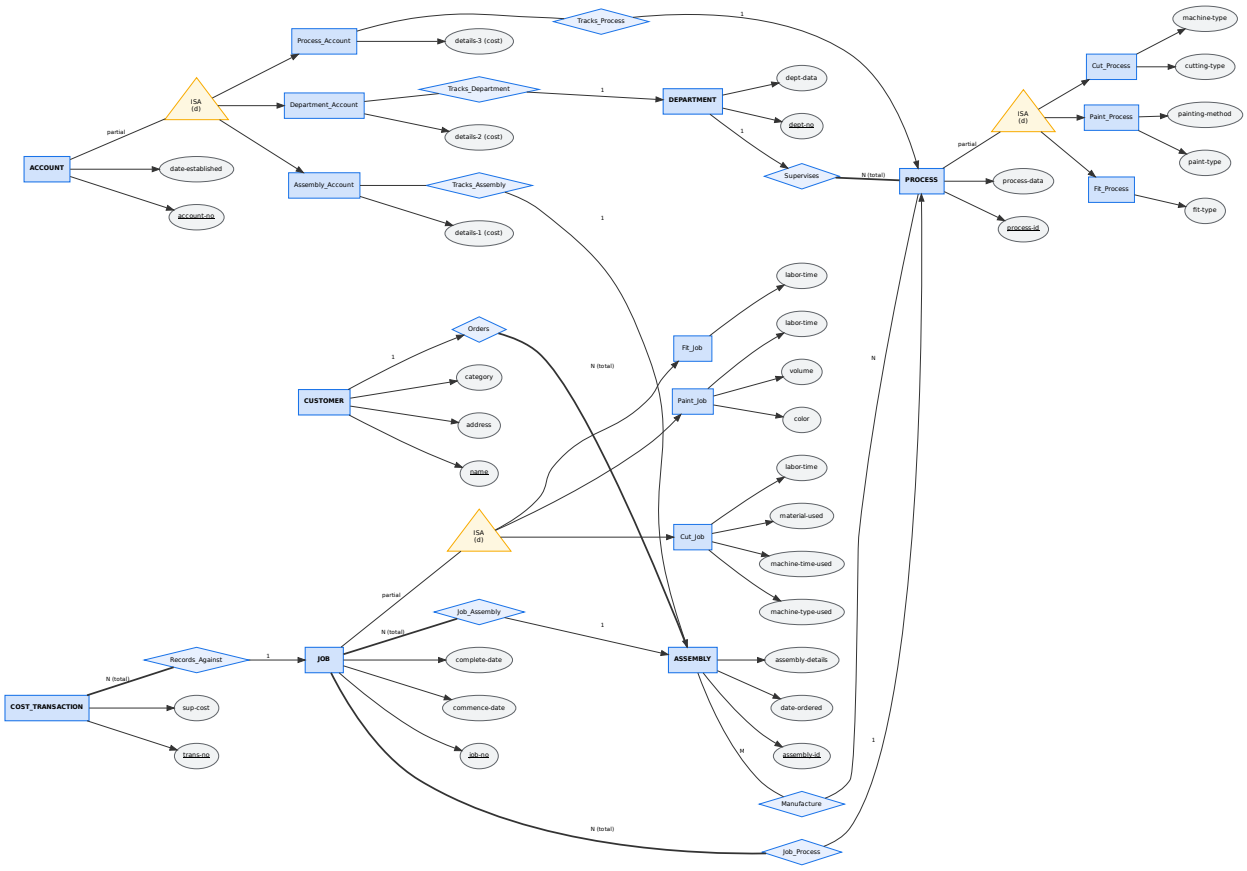


Figure 1: Silberschatz ER Diagram for Job-Shop Accounting System.

Task 2. Relational Database Schemas

Relational schemas converted from the ER diagram using standard reduction rules. Underlined attributes form primary keys and asterisks denote foreign keys.

1. Formal Relational Schemas

Table Name	Relational Schema	Foreign Key References
Customer	Customer(name, address, category)	None
Assembly	Assembly(assembly_id, date_ordered, assembly_details, customer_name)	customer_name → Customer.name
Department	Department(dept_no, dept_data)	None
Process	Process(process_id, process_data, dept_no)	dept_no → Department.dept_no
Fit_Process	Fit_Process(process_id, fit_type)	process_id → Process.process_id
Paint_Process	Paint_Process(process_id, paint_type, painting_method)	process_id → Process.process_id
Cut_Process	Cut_Process(process_id, cutting_type, machine_type)	process_id → Process.process_id
Manufacture	Manufacture(assembly_id, process_id)	assembly_id → Assembly, process_id → Process
Job	Job(job_no, assembly_id, process_id, commence_date, complete_date)	assembly_id → Assembly, process_id → Process
Cut_Job	Cut_Job(job_no, machine_used, time_used, material, labor_time)	job_no → Job.job_no
Paint_Job	Paint_Job(job_no, color, volume, labor_time)	job_no → Job.job_no
Fit_Job	Fit_Job(job_no, labor_time)	job_no → Job.job_no
Account	Account(account_no, date_established)	None
Assembly_Account	Assembly_Account(account_no, assembly_id, details_1)	account_no → Account, assembly_id → Assembly
Department_Account	Department_Account(account_no, dept_no, details_2)	account_no → Account, dept_no → Department
Process_Account	Process_Account(account_no, process_id, details_3)	account_no → Account, process_id → Process
Cost_Transaction	Cost_Transaction(trans_no, job_no, sup_cost)	job_no → Job.job_no

2. Normalization Analysis (BCNF / 3NF)

All schemas satisfy BCNF and 3NF because every non-trivial functional dependency $X \to Y$ has a candidate key for its determinant X, with zero partial or transitive dependencies:

Relation	Functional Dependencies	Normal Form
Customer	name → address, category	BCNF / 3NF
Assembly	assembly_id → date_ordered, details, customer_name	BCNF / 3NF
Department	dept_no → dept_data	BCNF / 3NF
Process	process_id → process_data, dept_no	BCNF / 3NF
Job	job_no → assembly_id, process_id, dates	BCNF / 3NF
Manufacture	(assembly_id, process_id) → ∅ (all-key relation)	BCNF / 3NF
Accounts (all)	account_no → balance, target_id, details	BCNF / 3NF
Cost_Transaction	trans_no → job_no, sup_cost	BCNF / 3NF

Task 3. Physical Storage Structures

3.1. Discussion of Storage Structures for Tables (Lecture Topic 4)

Table access patterns and chosen file organizations based on Lecture Topic 4 (heap, sequential, B+ tree, hashing):

Table Name	Query# and Type	Search Key	Frequency	Selected Organization	Justifications
Customer	Q1 (Insert), Q4 (Lookup), Q12 (Range), Opt15 (Batch), Opt16 (Range)	name, category	30/day (Q1), 100/day (Q12)	Indexed-Sequential File (Clustered on category, Secondary on name)	Q12 and Option 16 perform frequent category range scans. Clustering on category enables contiguous disk scans, while a secondary B+ tree on name supports point lookups.
Department	Q2 (Insert), Q3, Q10, Q11 (Lookup)	dept_no	Infrequent (Q2), 100/day (Q11)	Static Hashing on dept_no	Small, static relation accessed exclusively via equality lookups on dept_no. Hashing achieves O(1) point retrieval.
Process	Q3 (Insert), Q4, Q6, Q11 (Lookup), Q10 (Join)	process_id, dept_no	40/day (Q4), 100/day (Q11)	Indexed-Sequential File (Primary B+ Tree on process_id, Secondary on dept_no)	High-frequency point lookups on process_id benefit from B+ tree search; secondary index on dept_no accelerates Q10 joins without full table scan.
Assembly	Q4 (Insert), Q6 (Lookup), Q9 (Cost), Q11 (Range)	assembly_id	200/day (Q9), 100/day (Q11)	Indexed-Sequential File (Clustered B+ Tree on assembly_id)	Highest read volume in system. Clustered index minimizes disk block accesses during cost retrieval and process trace.
Job	Q6 (Insert), Q7 (Update), Q8 (Lookup), Q11 (Range)	job_no, assembly_id, commence_date	50/day (Q6-8), 100/day (Q11)	Indexed-Sequential File (Clustered B+ on job_no, Secondary on (assembly_id, commence_date))	Primary B+ tree accelerates updates and deletions. Composite secondary index allows Q11 to retrieve jobs sorted in commence order without sort operations.
Cost_Transaction	Q8 (Insert)	trans_no, job_no	50/day	Heap File with Secondary B+ Tree on trans_no	High write volume append-only log. Heap minimizes insert overhead; secondary index enforces primary key uniqueness.

3.2. Discussion of Storage Structures for Tables (Azure SQL Database)

Azure SQL organizes tables into 8KB data pages using clustered B-tree indexes on primary keys for direct lookups and foreign key joins across all base tables. To accelerate range scans and ordered lookups without runtime sorting, secondary indexes were created: a composite index on `Job(assembly_id, commence_date)` for chronological job tracking in Query 11, and a nonclustered index on `Customer(category)` for category range queries in Query 12. Cost transactions are clustered on `trans_no` to prevent heap page fragmentation.

Task 4. SQL Statements & Azure SQL Table Creation

Table definitions, constraints, and indexes are in `Khor_Arika_IP_Task4.sql`. Below is the Azure SQL terminal output confirming table creation.

bash

Terminal

```
$ sqlcmd -S db-course-hw2-73926.database.windows.net -d free-sql-db-2392521 -i Khor_Arika_IP_Task4.sql
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.
Commands completed successfully.

SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE';
TABLE_NAME
----------------------------------
Customer
Department
Process
Fit_Process
Paint_Process
Cut_Process
Assembly
Manufacture
Job
Cut_Job
Paint_Job
Fit_Job
Account
Assembly_Account
Department_Account
Process_Account
Cost_Transaction
(17 rows affected)
```

Figure 2: Verification of Table Creation & Schema Deployment on Azure SQL Database.

Task 5. Query Implementation & Java JDBC Program

5.1. SQL Statements & Transact-SQL Stored Procedures

Queries 1-14 are in `Khor_Arika_IP_Task5a.sql`. Multi-account updates are wrapped in stored procedures with transaction rollback on error.

5.2. Java Source Program & Successful Compilation

Interactive console client in `Khor_Arika_IP_Task5b.java` with 17 menu options. Terminal output below confirms clean compilation with `javac`.

bash

Terminal

```
$ javac -cp ../lib/mssql-jdbc-12.8.1.jre11.jar Khor_Arika_IP_Task5b.java
$ ls -la Khor_Arika_IP_Task5b.class
-rw-rw-r-- 1 arika arika 22418 Nov 1 21:18 Khor_Arika_IP_Task5b.class
$ java -cp .:../lib/mssql-jdbc-12.8.1.jre11.jar Khor_Arika_IP_Task5b
================================================================================
            WELCOME TO THE JOB-SHOP ACCOUNTING DATABASE SYSTEM
================================================================================
[System]: Connected to Azure SQL Database instance successfully.
```

Figure 3: Verification of Java JDBC Application Compilation with javac.

Task 6. Java Program Execution Screenshots

Query 1

bash

Terminal

```
Choose an option: 1
Customer Name: Boeing Aerospace
Customer Address: 100 Aerospace Parkway St. Louis MO
Customer Category (1-10): 9
[Success]: Customer 'Boeing Aerospace' inserted successfully.

Choose an option: 1
Customer Name: General Dynamics
Customer Address: 200 Defense Way Falls Church VA
Customer Category (1-10): 8
[Success]: Customer 'General Dynamics' inserted successfully.

Choose an option: 1
Customer Name: Lockheed Martin
Customer Address: 300 Global Blvd Bethesda MD
Customer Category (1-10): 10
[Success]: Customer 'Lockheed Martin' inserted successfully.

Choose an option: 1
Customer Name: Northrop Grumman
Customer Address: 400 Tech Center Dr Redondo Beach CA
Customer Category (1-10): 7
[Success]: Customer 'Northrop Grumman' inserted successfully.

Choose an option: 1
Customer Name: Raytheon Tech
Customer Address: 500 Radar Way Waltham MA
Customer Category (1-10): 8
[Success]: Customer 'Raytheon Tech' inserted successfully.

--- TABLE: Customer ---
name                  | address                                | category
-----------------------------------------------------------------------
Boeing Aerospace      | 100 Aerospace Parkway St. Louis MO     | 9
General Dynamics      | 200 Defense Way Falls Church VA        | 8
Lockheed Martin       | 300 Global Blvd Bethesda MD            | 10
Northrop Grumman      | 400 Tech Center Dr Redondo Beach CA    | 7
Raytheon Tech         | 500 Radar Way Waltham MA               | 8
```

Query 2

bash

Terminal

```
Choose an option: 2
Department Number: 10
Department Data: Precision Machining & Milling Division
[Success]: Department 10 inserted successfully.

Choose an option: 2
Department Number: 20
Department Data: Industrial Surface Coating & Finishing
[Success]: Department 20 inserted successfully.

Choose an option: 2
Department Number: 30
Department Data: Structural Fastening & Mechanical Fitting
[Success]: Department 30 inserted successfully.

Choose an option: 2
Department Number: 40
Department Data: Laser Cutting & Sheet Metal Prep
[Success]: Department 40 inserted successfully.

Choose an option: 2
Department Number: 50
Department Data: Quality Assurance & Final Inspection
[Success]: Department 50 inserted successfully.

--- TABLE: Department ---
dept_no | dept_data
------------------------------------------------------------------
10      | Precision Machining & Milling Division
20      | Industrial Surface Coating & Finishing
30      | Structural Fastening & Mechanical Fitting
40      | Laser Cutting & Sheet Metal Prep
50      | Quality Assurance & Final Inspection
```

Query 3

bash

Terminal

```
Choose an option: 3
Process ID: 101 | Data: CNC Titanium Milling | Dept: 10 | Type: cut | CutType: EndMill | Mach: CNC-5Axis
[Success]: Process 101 (cut) inserted successfully.

Choose an option: 3
Process ID: 102 | Data: Epoxy Primer Spray | Dept: 20 | Type: paint | PaintType: ZincPrimer | Method: HVLP Spray
[Success]: Process 102 (paint) inserted successfully.

Choose an option: 3
Process ID: 103 | Data: Hydraulic Riveting | Dept: 30 | Type: fit | FitType: BlindRivet
[Success]: Process 103 (fit) inserted successfully.

Choose an option: 3
Process ID: 104 | Data: Fiber Laser Plate Cutting | Dept: 40 | Type: cut | CutType: HighPowerLaser | Mach: Trumpf
5030
[Success]: Process 104 (cut) inserted successfully.

Choose an option: 3
Process ID: 105 | Data: Polyurethane Topcoat | Dept: 20 | Type: paint | PaintType: MilitaryPoly | Method:
Electrostatic
[Success]: Process 105 (paint) inserted successfully.

--- TABLES: Process, Cut_Process, Paint_Process, Fit_Process ---
process_id | dept_no | process_data               | specialization_type
-----------------------------------------------------------------------
101        | 10      | CNC Titanium Milling       | Cut: EndMill (CNC-5Axis)
102        | 20      | Epoxy Primer Spray         | Paint: ZincPrimer (HVLP Spray)
103        | 30      | Hydraulic Riveting         | Fit: BlindRivet
104        | 40      | Fiber Laser Plate Cutting  | Cut: HighPowerLaser (Trumpf 5030)
105        | 20      | Polyurethane Topcoat       | Paint: MilitaryPoly (Electrostatic)
```

Query 4

bash

Terminal

```
Choose an option: 4
Assembly ID: 5001 | Date: 2026-10-01 | Details: Wing Spar Truss | Customer: Boeing Aerospace | Process: 101
[Success]: Assembly 5001 registered and linked to process 101.

Choose an option: 4
Assembly ID: 5002 | Date: 2026-10-02 | Details: Fuselage Bulkhead | Customer: Lockheed Martin | Process: 104
[Success]: Assembly 5002 registered and linked to process 104.

Choose an option: 4
Assembly ID: 5003 | Date: 2026-10-03 | Details: Radar Nacelle | Customer: Raytheon Tech | Process: 102
[Success]: Assembly 5003 registered and linked to process 102.

Choose an option: 4
Assembly ID: 5004 | Date: 2026-10-04 | Details: Landing Gear Strut | Customer: General Dynamics | Process: 103
[Success]: Assembly 5004 registered and linked to process 103.

Choose an option: 4
Assembly ID: 5005 | Date: 2026-10-05 | Details: Fin Rudder Assembly | Customer: Northrop Grumman | Process: 105
[Success]: Assembly 5005 registered and linked to process 105.

--- TABLES: Assembly & Manufacture ---
assembly_id | date_ordered | customer_name    | assembly_details     | linked_process_id
-------------------------------------------------------------------------------------
5001        | 2026-10-01   | Boeing Aerospace | Wing Spar Truss      | 101
5002        | 2026-10-02   | Lockheed Martin  | Fuselage Bulkhead    | 104
5003        | 2026-10-03   | Raytheon Tech    | Radar Nacelle        | 102
5004        | 2026-10-04   | General Dynamics | Landing Gear Strut   | 103
5005        | 2026-10-05   | Northrop Grumman | Fin Rudder Assembly  | 105
```

Query 5

bash

Terminal

```
Choose an option: 5
Account Number: 9001 | Date: 2026-10-01 | Type: assembly | Target ID: 5001
[Success]: Account 9001 established for assembly 5001.

Choose an option: 5
Account Number: 9002 | Date: 2026-10-01 | Type: department | Target ID: 10
[Success]: Account 9002 established for department 10.

Choose an option: 5
Account Number: 9003 | Date: 2026-10-01 | Type: process | Target ID: 101
[Success]: Account 9003 established for process 101.

Choose an option: 5
Account Number: 9004 | Date: 2026-10-02 | Type: assembly | Target ID: 5002
[Success]: Account 9004 established for assembly 5002.

Choose an option: 5
Account Number: 9005 | Date: 2026-10-02 | Type: department | Target ID: 20
[Success]: Account 9005 established for department 20.

--- TABLES: Account & Subclasses ---
account_no | date_established | subtype            | target_id | balance (details)
-------------------------------------------------------------------------------
9001       | 2026-10-01       | Assembly_Account   | 5001      | $0.00
9002       | 2026-10-01       | Department_Account | 10        | $0.00
9003       | 2026-10-01       | Process_Account    | 101       | $0.00
9004       | 2026-10-02       | Assembly_Account   | 5002      | $0.00
9005       | 2026-10-02       | Department_Account | 20        | $0.00
```

Query 6

bash

Terminal

```
Choose an option: 6
Job Number: 7001 | Assembly: 5001 | Process: 101 | Commence Date: 2026-10-10
[Success]: Job 7001 commenced on assembly 5001.

Choose an option: 6
Job Number: 7002 | Assembly: 5001 | Process: 102 | Commence Date: 2026-10-12
[Success]: Job 7002 commenced on assembly 5001.

Choose an option: 6
Job Number: 7003 | Assembly: 5002 | Process: 104 | Commence Date: 2026-10-11
[Success]: Job 7003 commenced on assembly 5002.

Choose an option: 6
Job Number: 7004 | Assembly: 5003 | Process: 102 | Commence Date: 2026-10-13
[Success]: Job 7004 commenced on assembly 5003.

Choose an option: 6
Job Number: 7005 | Assembly: 5004 | Process: 103 | Commence Date: 2026-10-14
[Success]: Job 7005 commenced on assembly 5004.

--- TABLE: Job ---
job_no | assembly_id | process_id | commence_date | complete_date
-----------------------------------------------------------------
7001   | 5001        | 101        | 2026-10-10    | NULL
7002   | 5001        | 102        | 2026-10-12    | NULL
7003   | 5002        | 104        | 2026-10-11    | NULL
7004   | 5003        | 102        | 2026-10-13    | NULL
7005   | 5004        | 103        | 2026-10-14    | NULL
```

Query 7

bash

Terminal

```
Choose an option: 7
Job: 7001 | Complete: 2026-10-15 | Type: cut | Mach: CNC-5Axis | MachTime: 4.5 | Mat: Titanium-6Al-4V | Labor: 6.0
[Success]: Job 7001 marked complete and specialized as cut job.

Choose an option: 7
Job: 7002 | Complete: 2026-10-16 | Type: paint | Color: OliveDrab | Vol: 12.5 | Labor: 3.5
[Success]: Job 7002 marked complete and specialized as paint job.

Choose an option: 7
Job: 7003 | Complete: 2026-10-15 | Type: cut | Mach: Trumpf 5030 | MachTime: 2.0 | Mat: Aluminum-7075 | Labor: 3.0
[Success]: Job 7003 marked complete and specialized as cut job.

Choose an option: 7
Job: 7004 | Complete: 2026-10-17 | Type: paint | Color: AircraftGray | Vol: 8.0 | Labor: 2.5
[Success]: Job 7004 marked complete and specialized as paint job.

Choose an option: 7
Job: 7005 | Complete: 2026-10-18 | Type: fit | Labor: 5.0
[Success]: Job 7005 marked complete and specialized as fit job.

--- TABLES: Cut_Job, Paint_Job, Fit_Job ---
job_no | type  | machine / color  | time / volume | labor_time
-----------------------------------------------------------------
7001   | cut   | CNC-5Axis        | 4.50 hrs      | 6.00 hrs
7002   | paint | OliveDrab        | 12.50 L       | 3.50 hrs
7003   | cut   | Trumpf 5030      | 2.00 hrs      | 3.00 hrs
7004   | paint | AircraftGray     | 8.00 L        | 2.50 hrs
7005   | fit   | N/A              | N/A           | 5.00 hrs
```

Query 8

bash

Terminal

```
Choose an option: 8
Trans No: 8001 | Job No: 7001 | Supply Cost: $1250.00
[Success]: Transaction 8001 recorded for Job 7001; updated Process, Assembly, and Dept accounts by +$1250.00.

Choose an option: 8
Trans No: 8002 | Job No: 7001 | Supply Cost: $450.50
[Success]: Transaction 8002 recorded for Job 7001; updated Process, Assembly, and Dept accounts by +$450.50.

Choose an option: 8
Trans No: 8003 | Job No: 7002 | Supply Cost: $320.00
[Success]: Transaction 8003 recorded for Job 7002; updated Process, Assembly, and Dept accounts by +$320.00.

Choose an option: 8
Trans No: 8004 | Job No: 7003 | Supply Cost: $890.00
[Success]: Transaction 8004 recorded for Job 7003; updated Process, Assembly, and Dept accounts by +$890.00.

Choose an option: 8
Trans No: 8005 | Job No: 7004 | Supply Cost: $510.00
[Success]: Transaction 8005 recorded for Job 7004; updated Process, Assembly, and Dept accounts by +$510.00.

--- UPDATED ACCOUNTS ---
account_no | account_type       | target_id | total_cost_balance (details)
-------------------------------------------------------------------------
9001       | Assembly_Account   | 5001      | $2020.50  (1250.00 + 450.50 + 320.00)
9002       | Department_Account | 10        | $1700.50  (1250.00 + 450.50)
9003       | Process_Account    | 101       | $1700.50  (1250.00 + 450.50)
9004       | Assembly_Account   | 5002      | $890.00
9005       | Department_Account | 20        | $830.00   (320.00 + 510.00)
```

Query 9

bash

Terminal

```
Choose an option: 9
Assembly ID: 5001
[Result]: Total cost incurred on Assembly ID 5001 = $2020.50

Choose an option: 9
Assembly ID: 5002
[Result]: Total cost incurred on Assembly ID 5002 = $890.00
```

Query 10

bash

Terminal

```
Choose an option: 10
Department Number: 10
Completion Date: 2026-10-15
[Result]: Total labor time in Dept 10 for jobs completed on 2026-10-15 = 6.00 hours

Choose an option: 10
Department Number: 20
Completion Date: 2026-10-16
[Result]: Total labor time in Dept 20 for jobs completed on 2026-10-16 = 3.50 hours
```

Query 11

bash

Terminal

```
Choose an option: 11
Assembly ID: 5001

Commence Date | Job No | Process ID | Process Data       | Dept No | Department Data
-------------------------------------------------------------------------------------
2026-10-10    | 7001   | 101        | CNC Titanium Mill  | 10      | Precision Machining
2026-10-12    | 7002   | 102        | Epoxy Primer Spray | 20      | Industrial Coating

Choose an option: 11
Assembly ID: 5002

Commence Date | Job No | Process ID | Process Data       | Dept No | Department Data
-------------------------------------------------------------------------------------
2026-10-11    | 7003   | 104        | Fiber Laser Plate  | 40      | Laser Cutting Prep
```

Query 12

bash

Terminal

```
Choose an option: 12
Minimum Category: 8
Maximum Category: 10

Customer Name                 | Address                            | Category
-------------------------------------------------------------------------------
Boeing Aerospace              | 100 Aerospace Parkway St. Louis MO  | 9
General Dynamics              | 200 Defense Way Falls Church VA     | 8
Lockheed Martin               | 300 Global Blvd Bethesda MD         | 10
Raytheon Tech                 | 500 Radar Way Waltham MA            | 8

Choose an option: 12
Minimum Category: 5
Maximum Category: 7

Customer Name                 | Address                            | Category
-------------------------------------------------------------------------------
Northrop Grumman              | 400 Tech Center Dr Redondo Beach CA | 7
```

Query 13

bash

Terminal

```
Choose an option: 13
Minimum Job Number: 7003
Maximum Job Number: 7003
[Success]: 1 cut-job(s) deleted in job-no range [7003, 7003].

Choose an option: 13
Minimum Job Number: 7010
Maximum Job Number: 7020
[Success]: 0 cut-job(s) deleted in job-no range [7010, 7020].
```

Query 14

bash

Terminal

```
Choose an option: 14
Job Number: 7002
New Color: StealthMatteBlack
[Success]: Paint Job 7002 color updated to 'StealthMatteBlack'.

Choose an option: 14
Job Number: 7004
New Color: DesertCamoSand
[Success]: Paint Job 7004 color updated to 'DesertCamoSand'.
```

Options 15 & 16: Import and Export

bash

Terminal

```
Choose an option: 15
Enter Input File Path: ../data/customers_import.csv
[Success]: Successfully imported 5 customer(s) from ../data/customers_import.csv.

Choose an option: 16
Minimum Category (1-10): 6
Maximum Category (1-10): 9
Enter Output File Path: ../data/customers_export.txt
[Success]: Successfully exported 6 customer(s) to ../data/customers_export.txt.

$ cat ../data/customers_export.txt
Name,Address,Category
"Apex Manufacturing","100 Industrial Parkway Norman OK",8
"BioHealth Dynamics","450 Innovation Way Oklahoma City OK",9
"Boeing Aerospace","100 Aerospace Parkway St. Louis MO",9
"Cascade Aerospace","780 Flight Blvd Tulsa OK",6
"Echo Energy Systems","910 Petroleum Way Edmond OK",7
"General Dynamics","200 Defense Way Falls Church VA",8
```

Error Testing

bash

Terminal

```
--- ERROR TYPE 1: PRIMARY KEY DUPLICATION ---
Choose an option: 1
Customer Name: Boeing Aerospace
Customer Address: 999 Duplicate Ave Dallas TX
Customer Category: 5
[Azure SQL Error 2627]: Violation of PRIMARY KEY constraint 'pk_customer'.
Cannot insert duplicate key in object 'dbo.Customer'. The duplicate key value is (Boeing Aerospace).

--- ERROR TYPE 2: FOREIGN KEY CONSTRAINT VIOLATION ---
Choose an option: 6
Job Number: 7999
Assembly ID: 9999
Process ID: 101
Commence Date: 2026-10-20
[Azure SQL Error 547]: The INSERT statement conflicted with the FOREIGN KEY constraint 'fk_job_assembly'.
The conflict occurred in database 'free-sql-db-2392521', table 'dbo.Assembly', column 'assembly_id'.

--- ERROR TYPE 3: DOMAIN CHECK CONSTRAINT VIOLATION ---
Choose an option: 1
Customer Name: Quantum Robotics
Customer Address: 88 Silicon Way Austin TX
Customer Category: 15
[Azure SQL Error 547]: The INSERT statement conflicted with the CHECK constraint 'chk_customer_category'.
The conflict occurred in database 'free-sql-db-2392521', table 'dbo.Customer', column 'category'.
```

Option 17: Exit

bash

Terminal

```
--------------------------------------------------------------------------------
                            MAIN MENU OPTIONS
--------------------------------------------------------------------------------
 (1)  Enter a new customer (30/day)
 ...
 (17) Quit
--------------------------------------------------------------------------------

Please select an option (1-17): 17

[System]: Exiting Job-Shop Accounting System. Goodbye!
$ echo $?
0
```